ELECTRODYNAMICS II · SEMESTER 5

Fields in matter

Module 1 · PYQ + model paper

40 questions · 20 short + 16 problems + 4 essays
Double the question count of the supplied April 2026 FYUGP prerequisite paper.

CU-FYUGP syllabus 2024 · study edition
SI units · phone pages · October 2026

Practice instructions

Suggested time: two sittings of two hours. Section A: 20 questions × 3 marks, ceiling 48. Section B: 16 × 6, ceiling 72. Section C: answer any two of four × 10. Maximum: 140.

This doubles the supplied April 2026 **Electrodynamics I** 70-mark practice pattern. The current ED II syllabus specifies 70 theory marks but does not supply a question pattern here. This is a study paper, not an official FYUGP mock.

Each numbered question has a distinct primary assessment target. Longer questions combine earlier methods. N25 means November 2025 CBCSS ED I; A26 means April 2026 FYUGP ED I. Tags explicitly identify these older or prerequisite courses. Expanded/adapted means wording, scope or mark allocation was changed. Overlapping past questions are tagged with their original course and number; untagged questions are original model practice.

All four theory modules are included in the set. Module V is laboratory work and is outside this theory-paper series.

Section A · Short answers

3 marks each · ceiling 48

1. How can an electrically neutral dielectric respond to a static electric field?

Model · syllabus-based

2. Define atomic polarizability and give its SI unit.

Legacy PYQ N25 Q5 · expanded

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3. A permanent dipole has moment 2×10^{-29} C m in a field 3×10^{5} V/m at 30°. Find the torque magnitude.

Model · syllabus-based

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4. Which orientation of a permanent dipole in a uniform field is stable? Use energy.

Model · syllabus-based

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5. Define polarization and distinguish it from dipole moment.

Model · syllabus-based

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6. A material has 5×10^{27} aligned dipoles per m^3, each of moment 2×10^{-30} C m. Find P.

Model · syllabus-based

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7. A face normal makes 60° with uniform $\mathbf{P}$, whose magnitude is 0.04 C/m². Find the bound surface charge density.

Model · syllabus-based

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8. What is the physical interpretation of bound charges? Include the case of uniform polarization.

Legacy PYQ N25 Q10 · expanded

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9. Define electric displacement and state its SI unit.

Model · syllabus-based

10. Which charge appears in Gauss's law for D, and which appears in Gauss's law for E?

Model · syllabus-based

11. State the boundary condition for the normal component of D, with your normal convention.

Model · syllabus-based

12. Why is tangential E continuous across an electrostatic interface?

Model · syllabus-based

13. A linear dielectric has $\epsilon_r = 5$. Find its electric susceptibility.

Model · syllabus-based

14. A capacitor stays connected to an ideal battery while a dielectric completely fills it. How do capacitance and free charge change?

Model · syllabus-based

15. Distinguish paramagnets and diamagnets.

Legacy PYQ N25 Q9

16. Why does a paramagnet have weak alignment rather than perfect alignment at ordinary temperature?

Model · syllabus-based

17. Define magnetic vector potential. How can it describe the field of a magnetized object?

Legacy PYQ N25 Q11 · expanded

18. Discuss briefly the bound currents in a magnetized medium, emphasizing their physical interpretation.

Legacy PYQ N25 Q18 · adapted marks

19. Define H and state its magnetostatic circulation law.

Model · syllabus-based

20. Define magnetic susceptibility and relate it to relative permeability for a linear isotropic material.

Legacy PYQ N25 Q3 · expanded

Section B · Problems and explanations

6 marks each · ceiling 72

21. Explain remanence, coercivity and the role of domain behaviour in an M-H hysteresis loop.

Model · syllabus-based

22. Explain the effect of heating a ferromagnet through the Curie temperature, and distinguish the result from diamagnetism.

Model · syllabus-based

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23. Model an atom as a point nucleus +q in a uniform spherical cloud −q of radius a. Derive its polarizability for a small nucleus displacement d.

Model · syllabus-based

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24. A fixed dipole $\mathbf{p} = p\hat{\mathbf{x}}$ is in $\mathbf{E} = kx^2\hat{\mathbf{x}}$. Find its force at x. Is there a torque?

Model · syllabus-based

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25. A sphere of radius R has $\mathbf{P} = a\mathbf{r}$ inside and zero outside. Find its bound charge densities and total bound charge.

Model · syllabus-based

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26. An infinite slab is uniformly polarized perpendicular to its faces, with no free charge. Find E inside and outside.

Model · syllabus-based

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27. A long straight wire with uniform free line charge λ is surrounded by rubber insulation out to radius a. Find the electric displacement, and state E if the rubber has permittivity ε and the exterior is vacuum.

Legacy PYQ N25 Q19 · expanded

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28. At an uncharged interface between isotropic dielectrics, derive the relation between field angles θ_1, θ_2 measured from the normal.

Model · syllabus-based

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29. A fully filled parallel-plate capacitor has area 0.010 m², separation 1 mm and $\epsilon_r = 4$. At 100 V find C, free Q and P.

Model · syllabus-based

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30. A charged capacitor is disconnected from its battery before a dielectric of relative permittivity k is inserted fully. Derive the ratios of V and stored energy.

Model · syllabus-based

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31. Obtain the torque expression for a planar current loop. Then consider a 100-turn coil of area 2×10^{-3} m² carrying 0.20 A. Its normal makes 30° with B=0.50 T. Find moment, torque and dipole energy.

Prerequisite-course PYQ A26 Q10 · expanded

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32. A very long cylinder has uniform $\mathbf{M} = M\hat{\mathbf{z}}$ and no free current. Find its bound currents and B far from its ends.

Model · syllabus-based

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33. Inside a block, $\mathbf{M} = ax\hat{\mathbf{z}}$. Find its volume bound current and the surface current on the face x=L.

Model · syllabus-based

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34. A long solenoid has 1000 turns/m, current 0.50 A and a linear core with $\mu_r = 2$. Find H, M and B inside, and state the outside-field approximation. Recover the air-core case.

Prerequisite-course PYQ A26 Q16 · adapted to matter

35. Compare P, D, M and H, and explain why $D = \epsilon_0 E$ and $B = \mu_0 H$ are not general material identities.

Model · syllabus-based

36. Explain why a transformer core and a permanent magnet require different hysteresis behaviour.

Model · syllabus-based

Section C · Essays

10 marks each · answer any two

37. Starting from dipole potential, derive the surface and volume bound-charge representation of a polarized object and explain its physical meaning.

Legacy PYQ N25 Q17 · adapted marks

38. Develop a systematic method for solving a planar dielectric interface problem with free surface charge. Include the field laws and a normal-field example.

Model · syllabus-based

39. Explain how microscopic dipoles lead to M, bound currents and H. Derive the free-current Ampère law and discuss its limits.

Model · syllabus-based

40. Compare diamagnetism, paramagnetism and ferromagnetism, including microscopic origin, field response and hysteresis.

Model · syllabus-based

Coverage audit

40 numbered questions; 40 primary targets. The list below makes the intended distinction explicit.

Q1 · **T1** Insulator response
Q2 · **T1** Atomic polarizability
Q3 · **T2** Dipole torque
Q4 · **T2** Stable orientation
Q5 · **T3** Polarization definition
Q6 · **T3** Number density conversion
Q7 · **T4** Surface bound charge projection
Q8 · **T4** Physical interpretation of bound charge
Q9 · **T5** Definition of D
Q10 · **T5** Free versus bound source
Q11 · **T6** Normal electric discontinuity
Q12 · **T6** Tangential electric continuity
Q13 · **T7** Relative permittivity relation
Q14 · **T7** Fixed-voltage insertion
Q15 · **T8** Diamagnetic versus paramagnetic response
Q16 · **T8** Paramagnetic temperature competition
Q17 · **T10** Magnetic vector potential
Q18 · **T10** Cancellation of microscopic loops
Q19 · **T11** Definition of H
Q20 · **T12** Magnetic susceptibility definition

Q21 · **T13** Remanence and coercivity

Q22 · **T13** Curie transition

Q23 · **T1** Classical atomic model

Q24 · **T2** Nonuniform dipole force

Q25 · **T4** Bound charge conservation

Q26 · **T4** Depolarizing field of a slab

Q27 · **T5** Line free charge in a dielectric

Q28 · **T6** Refraction of electrostatic field lines

Q29 · **T7** Capacitance calculation

Q30 · **T7** Isolated-capacitor energy

Q31 · **T9** Magnetic dipole torque calculation

Q32 · **T10** Uniformly magnetized cylinder

Q33 · **T10** Nonuniform magnetization curl

Q34 · **T11** Solenoid with magnetic filling

Q35 · **T12** Comparing electric and magnetic auxiliaries

Q36 · **T13** Choice of magnetic material

Q37 · **T4** Derivation of equivalent bound charges

Q38 · **T6** Dielectric boundary-value method

Q39 · **T11** Macroscopic magnetostatic description

Q40 · **T13** Magnetic-material classification

Past-paper accounting

10 distinct questions from the supplied ED I papers are represented at Q2, Q8, Q15, Q17, Q18, Q20, Q27, Q31, Q34, Q37. Their course, date and any adaptations are printed beside the questions. Full original wordings and exclusions are in the source inventory.